Week 6 – Weekly Homework

Library Loan Report

Use the provided libraryData to generate a library loan report and prepare independent reminder requests.

Git is required

Part 1 – Build Loan Rows

Write a function named buildLoanRows(data).

The function must return a new array containing one object for each loan, in the same order as the original loans array.

Property	Description
loanId	Loan ID
bookTitle	Matching book title
readerLabel	Reader display name, or "عضو ناشناس" if not found
emailLabel	Reader email, or "ایمیل ثبت نشده" if missing or empty
status	"returned", "on-time", or "overdue"
remainingDays	Number of days remaining
overdueDays	Number of overdue days

Status rules:

- If returned is true, status is "returned" and both day values are 0.
- If the loan is active and elapsedDays <= loanDays, status is "on-time". Calculate remainingDays and set overdueDays to 0.
- If the loan is active and elapsedDays > loanDays, status is "overdue". Calculate overdueDays and set remainingDays to 0.

Requirements:

- Use map() to create the rows and find() to locate books and readers.
- Use named callbacks and simple destructuring for at least two properties.
- Use optional chaining when accessing contact information.
- Do not modify the original data.
- Return an empty array if there are no loans.

Part 2 – Build the Library Report

Write a function named buildLibraryReport(data).

It must return a new object containing exactly these properties:

Property	Description
loanCount	Total number of loans
activeCount	Number of loans that have not been returned
overdueCount	Number of active overdue loans
totalOverdueDays	Total overdue days of active loans
unmatchedReaderLoanCount	Number of loans with no matching reader
mostOverdueLoanId	ID of the active loan with the highest overdue days, or null if none
byReader	A report for every reader

Each object inside byReader must contain exactly:

- readerId
- readerLabel
- activeCount
- overdueCount
- totalOverdueDays

Include every reader, even those with no loans. Only count loans that match that reader's ID. Unknown readers must not appear in byReader.

Requirements:

- You may reuse buildLoanRows() from Part 1.
- Use filter() to select active and overdue loans.
- Use a loop or reduce() with an initial value of 0 for totals.
- Find the most overdue loan without using sort(). If there is a tie, keep the first loan in the original array.
- Do not use console.log() inside the function.
- Do not use global counters.

Expected overall results for the provided data:

Property	Expected value
loanCount	6
activeCount	5
overdueCount	3
totalOverdueDays	8
unmatchedReaderLoanCount	1
mostOverdueLoanId	"L1"

For empty data, all counts and totals must be 0, mostOverdueLoanId must be null, and byReader must be an empty array.

Part 3 – Prepare a Reminder Request

Write a function named prepareReminderRequest(data, loanId).

The function must return a new loan object or null.

A reminder request is allowed only when:

1. The loan exists.
2. The loan has not been returned.
3. The loan is overdue.
4. The reader exists.
5. The reader has a non-empty email.

If any condition is not met, return null.

For a valid request:

- Return a new loan object.
- Preserve all original loan properties.
- Create a new reminder object.
- Set reminder.requested to true.
- Preserve the original reminder.note.

Important:

- Do not modify the original data.

- The returned loan and its reminder object must not share references with the original objects.
- Only create the copies needed for this task. A general deep copy is not required.
- No actual email should be sent.

Expected results:

Loan ID	Expected result
L1	New loan object with requested: true
L2	null – no email
L3	null – not overdue
L4	null – already returned
L5	null – reader not found
L6	null – not overdue
L99	null – loan not found

After preparing the request for L1, the original loan's reminder.requested must remain false.